

Fruzsina Julia Agocs — Curriculum Vitae

Assistant Professor at the University of Colorado, Boulder, Department of Computer Science
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 arXiv

Research interests

Numerical analysis	Oscillatory problems
Numerical solution of PDEs	Wave propagation
Boundary integral equations	Computational cosmology, physics
Numerical solution of ODEs	Open-source software
High-order methods	

Education

Ph.D. Theoretical Cosmology, University of Cambridge, UK	2021
<i>Advisor: Anthony Lasenby, Mike Hobson, Will Handley</i>	
Thesis: Primordial evolution of cosmological perturbations: Theory and computation	
M.Sci. Theoretical and Experimental Physics, University of Cambridge, UK	2017
<i>Advisor: Will Handley, First class</i>	
Dissertation: The Runge-Kutta-Wentzel-Kramers-Brillouin method and the primordial Universe	
B.A. Natural Sciences, University of Cambridge, UK	2016
First class	

Professional experience

Assistant Professor, Department of Computer Science, CU Boulder	Aug 2024–
Flatiron Research Fellow, CCM, Flatiron Institute	2021–2024
Research student, British Antarctic Survey	Jul 2019– Jan 2020
<i>Forecasting Arctic sea ice extent with temporal convolutional networks</i>	
<i>Advisor: Scott Hosking</i>	
Research Engineer, Kokoon Technology Ltd.	Jul–Sept 2016 & Jul–Aug 2017
<i>Sleep cycle classification based on single-channel EEG data</i>	
Research student, Institute of Astronomy, University of Cambridge	Jun–Sept 2015
<i>Correcting for host galaxy contamination in the spectral energy distribution of active galactic nuclei</i>	
<i>Advisor: Ranjan Vasudevan</i>	

Publications

[Google scholar](#) 

Refereed: 11 articles (9 first author)

Citations 391

h-index: 7

(as of 2026-10-08)

Refereed

- 11 Ohana, R. et al. (incl. **FJA**), [The Well: a Large-Scale Collection of Diverse Physics Simulations for Machine Learning](#), Advances in Neural Information Processing Systems, **37**, 44989-45037, 2024 (arXiv:2412.00568), [215 citations]
- 10 **Agocs, F. J.**; Barnett, A. H., [Trapped acoustic waves and raindrops: High-order accurate integral equation method for localized excitation of a periodic staircase](#), Journal of Computational Physics, **519**, 113383, 2024 (arXiv:2310.12486) [6 citations]
- 9 Letey, M. I.; Shumaylov, Z.; **Agocs, F. J.**; Handley, W. J. et al., [Quantum initial conditions for curved inflating universes](#), Physical Review D, **109**, 123502, 2024 (arXiv:2211.17248) [19 citations]
- 8 **Agocs, F. J.**; Barnett, A. H., [An adaptive spectral method for oscillatory second-order linear ODEs with frequency-independent cost](#), SIAM Journal on Numerical Analysis, **62**, 295-321, 2024 (arXiv:2212.06924) [11 citations]
- 7 **Agocs, F. J.**; Barnett, A., [riccati: an adaptive, spectral solver for oscillatory ODEs](#), JOSS, **8**, 5430, 2023 [2 citations]
- 6 Hergt, L. T.; **Agocs, F. J.**; Handley, W. J.; Hobson, M. P. et al., [Finite inflation in curved space](#), Physical Review D, **106**, 63529, 2022 (arXiv:2205.07374) [28 citations]
- 5 AbdusSalam, S. S.; **Agocs, F. J.**; Allanach, B. C.; Athron, P. et al., [Simple and statistically sound recommendations for analysing physical theories](#), Reports on Progress in Physics, **85**, 52201, 2022 (arXiv:2012.09874) [35 citations]
- 4 **Agocs, F. J.**, [\(py\)oscode: fast solutions of oscillatory ODEs](#), JOSS, **5**, 2830, 2020 [4 citations]
- 3 **Agocs, F. J.**; Hergt, L. T.; Handley, W. J.; Lasenby, A. N. et al., [Quantum initial conditions for inflation and canonical invariance](#), Physical Review D, **102**, 23507, 2020 (arXiv:2002.07042) [14 citations]
- 2 Andersson, T.; **Agocs, F. J.**; Hosking, S.; Pérez-Ortiz, M. et al., [Deep learning for monthly Arctic sea ice concentration prediction](#), EGU General Assembly Conference Abstracts, 15481, 2020 [2 citations]
- 1 **Agocs, F. J.**; Handley, W. J.; Lasenby, A. N.; Hobson, M. P., [Efficient method for solving highly oscillatory ordinary differential equations with applications to physical systems](#), Physical Review Research, **2**, 13030, 2020 (arXiv:1906.01421) [46 citations]

Preprints & other

- 4 **Agocs, F. J.**; Goodwill, T.; Hoskins, J. G.; Nekrasov, P., [Integral equations for flexural scattering problems with periodic boundaries](#), 2026 (arXiv:2603.00366)
- 3 Fisher, R.; **Agocs, F. J.**; Gillman, A., [An efficient boundary integral equation solution technique for solving aperiodic scattering problems from two-dimensional, periodic boundaries](#), 2025 (arXiv:2511.04424) [1 citation]
- 2 **Agocs, F. J.**; Goodwill, T.; Hoskins, J., [Complex Scaling for the Junction of Semi-infinite Gratings](#), 2025 (arXiv:2509.02951) [3 citations]
- 1 **Agocs, F. J.**; Hobson, M. P.; Handley, W. J.; Lasenby, A. N., [Dense output for highly oscillatory numerical solutions](#), 2020 (arXiv:2007.05013) [5 citations]

Honors and awards

STFC funded PhD award	2017–2021
Duncan Bruce memorial prize for excellence in physics	2017
Senior scholarship for academic excellence	2016

Talks

Nov 2027, ICERM Applied and Computational Complex Analysis in the 21st Century: Frontiers and Future Perspectives (invited speaker)
 Jul 2027, ICOSAHOM, Milan (invited minisymposium)
 Feb 2027, SIAM CSE (invited minisymposium)
 Mar 2026, Banff Women in Mathematics workshop, Canada (invited participant)
 Jul 2025, ICOSAHOM, Montreal (invited minisymposium)
 Jun 2025, SDIDE 2025, Italy (invited talk)
 Mar 2025, SIAM CSE (invited minisymposium)
 Mar 2024, Harvard University (job talk)
 Mar 2024, University of Colorado, Boulder (job talk)
 Feb 2024, University of Maryland, Baltimore (job talk)
 Jan 2024, University of Warwick, UK (job talk)
 Nov 2023, University of Massachusetts Lowell (invited seminar)
 Oct 2023, SIAM-NNP, Newark (minisymposium, organizer)
 Oct 2023, Cornell (invited talk)
 Aug 2023, ICIAM, Tokyo (contributed talk)
 Aug 2023, ICOSAHOM, Seoul (minisymposium)
 Apr 2023, NYU, New York (invited seminar)
 Feb 2023, SIAM Computer Science and Engineering, Amsterdam (minisymposium, organizer)
 Feb 2023, Yale (invited seminar)
 Dec 2022, University of Innsbruck (invited seminar)
 Oct 2022, Flatiron-wide algorithms and mathematics ($F_w(\alpha + m)!$), New York (lecture)
 Oct 2022, University of Chicago (invited seminar)
 Oct 2022, New Jersey Institute of Technology (invited seminar)
 Sept 2022, Sayas numerics day, University of Maryland, BC (contributed talk)
 Jun 2022, SDIDE, Budapest (invited speaker)
 May 2022, BIRS-CMO workshop on "Outstanding challenges in computational methods for integral equations", virtual (invited speaker)
 Jan 2021, CAM-LMU workshop, virtual (contributed talk)
 Nov 2020, Numerical analysis seminar, Flatiron Institute, New York
 July 2020, Beecroft Institute, University of Oxford (seminar)
 July 2020, SciPy conference, virtual (contributed talk)

Apr 2020, Battcock Centre for Experimental Astrophysics, Cambridge (invited seminar)
 Jan 2020, Institute of Astronomy, Cambridge (invited seminar)
 Sept 2019, KICC10, Kavli Institute of Cosmology, Cambridge (contributed talk & poster)
 Oct 2018, Kavli Institute of Cosmology, Cambridge (invited seminar)
 Jul 2018, National CDT in data intensive science conference, London (poster)
 Jun 2017, Battcock Centre for Experimental Astrophysics, Cambridge (invited seminar)

Open source development

Core developer and maintainer of [riccaticpp](#), [riccati](#), [oscode](#)
 Member of “core team” of [GAMBIT](#)

Teaching

2026

Spring: CSCI 3656 “Numerical Computation”
 Fall: CSCI 5525 “[Computational tools for multiscale problems](#)”

2025

Spring: CSCI 5636 “[Numerical solution of partial differential equations](#)”
 Fall: CSCI 5526 “[Computational tools for multiscale problems](#)”
 PhD proposal defense/qualifying exam committee member for 3 students

2024

Fall: CSCI 7000-014 “[Computational tools for multiscale problems](#)”
 Fall: PhD qualifying exam committee member for 2 students

Earlier

Tutorials in groups of 2-20, University of Cambridge:
 2018–2019, 1st year Mathematics (*6 students, 56 hours total*)
 2017–2019, 3rd year General Relativity (*11 students, 11 hours total*)
 2017–2020, 4th year Relativistic Astrophysics and Cosmology (*50 students, 28 hours total*)

Mentorship

PhD

Riley Fisher (with Adrianna Gillman)	2025–
Stefano Fochesatto (with Jed Brown)	2025–

Undergraduate

Jorge Lopez Leon (summer student)	2026
<i>Fast solution of nonlinear oscillatory ODEs for physical systems</i>	
Winner of best poster prize, undergraduate research symposium	

Andy Gusty (senior thesis, with Emily Jensen, Cody Scarborough, and Elizabeth Bradley) 2025
Optimal design of nonlinear periodic electrical networks for soliton pulse generation
 Winner of undergraduate research award, NSF GRFP

Sankalan Bhattacharyya (co-mentor with Will Handley) 2023, University of Cambridge
Applications of spectral collocation methods in cosmology simulations

Public outreach (selected)

2020, [Public talk](#) at the Open Evening of the Institute of Astronomy, Cambridge
 2020, Astronomy lecture and observation session at [Logikatábor](#), Hungary
 2020, Interview for [article](#) on meteor showers
 2020, [Periscope broadcast](#) at RWTH Aachen University
 2018, [Invited speaker at the Women in STEM residential](#), Gonville and Caius college

Service

Journal editor

Journal of Open Source Software
 2026: 3 accepted, 3 in review
 2025: 2 accepted

Peer-review

Advances in Computational Mathematics (2)
 Journal of Computational and Applied Mathematics (1)
 Journal of Open Source Software (3)
 SIAM Numerical Analysis (1)
 SciPy conference (3)

Workshop and meeting organization

Oct 2023, SIAM regional meeting, minisymposium on "Advances in integral equations and quadrature methods, and their applications in computational physics"
 Feb 2023, SIAM Computer Science and Engineering, minisymposium on "Software for integral equations and boundary element methods"

Committees

Faculty search committee in quantum computing	2026
Undergraduate curriculum committee	2025–
Graduate committee	2024–2025